

Resistance-Aware Docking Stratifies Antimalarial Leads by Mutant Retention but Short MD Shows Non-Equivalent Estimands

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Abstract

Resistance-aware polypharmacology docking stratifies antimalarial candidates by predicted mutant retention, but whether docking-derived and molecular-dynamics-derived resistance scores measure the same quantity has not been tested. We computed a docking-derived resistance-retention score (RRS) on 136 PfDHFR and PfCRT wild-type and mutant systems for a 17-candidate African-natural-product cohort. The primary analysis on 12 two-target candidates produced a five-class stratification (A*: 1, A: 1, B: 4, C: 5, D: 1) that separates predicted wild-type potency from predicted mutant retention; network score and RRS were weakly associated ($\rho = -0.2098$, adjusted $p = 1.0000$), indicating that the three ranking dimensions capture distinct

information. Short 10 ns MD on four non-overlapping parent-study complexes left only PfCRT-214 with a bound ligand and an interpretable MM-GBSA estimate (-18.25 ± 0.40) kcal mol⁻¹). A 16-system single-replicate pilot on two Set-C candidates did not recover the docking retention pattern; trajectory metrics and docking-RRS disagreed in direction for seven of eight mutant comparisons, demonstrating that docking-RRS and MD-RRS are non-equivalent estimands. An external 39-ligand panel (312 records) gave a mean RRS near 100, confirming that docking-score retention is protocol-dependent and not a general property of the scoring function. Docking-based RRS stratifies candidates inside this computational panel, but the ranking is hypothesis-generating only; short MD stress tests probe a different estimand and cannot validate docking-derived retention classes.

Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden¹. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries², and declining efficacy of some partner drugs has been reported^{1,3}. Spatial-temporal modelling further indicates that the geographic footprint of *pfk13* markers in Africa is expanding⁴, while genomic surveillance documents the co-occurrence of *Pfcr*t and *Pfdhfr* resistance variants across diverse transmission settings⁵⁻⁷.

Polypharmacology is a design strategy for complex disease biology, and multi-target-directed ligands represent an important translational class⁸. The rationale is evolutionary as well as pharmacological: under the single-drug-multi-target (SDMT) paradigm, simultaneous loss of activity at several relevant targets raises the mutational barrier to treatment failure in a way that combination therapy alone does not⁹. That proposition is conditional, however. A list of predicted protein contacts is not a mechanism-of-action assignment; causal

network analyses require an explicit perturbation model, a defined biological background, and evidence that the predicted interactions propagate to a phenotype¹⁰. Network-learning approaches to off-target prediction illustrate the value of graph context while underscoring that network inference remains distinct from experimental target engagement¹¹. Natural products are attractive starting points because their stereochemical and scaffold complexity supports activity across multiple protein classes^{12,13}. African natural products provide a particularly relevant chemical-space context, with resources such as ANPDB enabling explicit comparison with approved-drug and synthetic reference spaces¹⁴.

In a previous study, we used generative molecular methods and dual-filter consensus docking (AutoDock Vina¹⁵ and DiffDock-L¹⁶) to generate a library of 65 856 candidate molecules from 396 African natural products and 454 synthetic drugs¹⁷. We selected the present candidates against a panel spanning established resistance determinants and mechanistically distinct parasite processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and the 4GM2 entry represents PfClpR, not PfClpP, and is retained as a proteostasis-related parent-study system. This panel is deliberately a mechanistic cross-section rather than an exhaustive parasite targetome. It allows us to ask whether resistance-aware prioritization remains informative across distinct target classes, while avoiding the stronger claim that the selected compounds define a complete parasite-wide polypharmacology profile. PfCRT docking and MD were performed at physiological pH; results for this transporter are therefore conditional on that preparation and may differ under digestive-vacuole pH.

Rigid docking is efficient for ranking but treats the receptor and scoring landscape in a simplified manner. MD tests whether a docked pose remains compatible with protein flexibility, whereas endpoint methods such as MM-GBSA provide system-specific energetic summaries rather than calibrated thermodynamic observables^{18,19}. This distinction is especially important for resistance analysis: a docking RRS is a hypothesis about relative

mutant sensitivity, not a measurement of a mutant free-energy difference. Network pharmacology provides a complementary prioritization layer, but centrality is a topological weighting heuristic rather than a causal measure of target essentiality, consistent with the growing interest in multi-target antimalarial design more broadly²⁰. We therefore treat these quantities separately: docking-derived RRS for mutant sensitivity, PNS for network-weighted target ranking, ACSI for chemical-space positioning, and targeted MD for post-docking structural scrutiny.

Workflows that combine a docking-derived retention metric, network pharmacology and short MM-GBSA endpoints are sometimes presented as joint evidence for resistance-resilient polypharmacology, but each layer measures a different estimand and has its own scope of inference. Treating them as independent evidence of mechanism, rather than as parallel ranking signals with non-overlapping failure modes, inflates the apparent epistemic reach of a single short MD pilot. We make this distinction explicit throughout.

The central question is therefore twofold: (i) whether resistance-aware polypharmacology docking produces a discriminant prioritization that separates candidates by their predicted retention across clinically relevant mutant panels, and (ii) whether short molecular dynamics simulations corroborate the docking-derived retention signal or refute it by revealing that the two approaches measure non-equivalent biophysical quantities. The first question addresses whether RRS-based stratification is informative enough to generate hypotheses for downstream experimental testing. The second addresses whether MD, despite its richer representation of protein flexibility, converges on the same compound ranking—or instead exposes a fundamental mismatch between static docking scores and dynamic binding behavior. The primary analysis used 12 candidates with eligible PfDHFR and PfCRT wild-type scores; five PfCRT-only candidates formed a coverage-limited sensitivity set. RRS was calculated from empirical Vina scores, whereas PNS and ACSI provided complementary ranking dimensions. Four parent-study wild-type complexes—chemically and numerically distinct from the 17 Set-C candidates—underwent 10 ns simulations as a methodological pose-retention check; a

16-system Set-C pilot provided the only trajectory-level comparison against docking-RRS for members of the prioritization cohort. Biological polypharmacology and resistance circumvention were not tested.

Methods

Study design

The primary analysis is docking-score retention (RRS) for the 12 Set-C candidates that have eligible PfDHFR and PfCRT wild-type scores. The central estimand is a computational triage signal, not a biological resilience phenotype. We therefore distinguish the primary docking-RRS analysis from secondary structural-retention diagnostics, technical validation, and unobserved biological quantities. Three secondary analyses are reported separately: the five PfCRT-only candidates, a four-complex parent-study MD check, and a 16-system Set-C MD pilot. They address different coverage and sampling questions and are not combined with the primary RRS classes. The parent-study MD systems are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Candidate selection and filtering

The Set-C cohort is a filtered prioritization cohort rather than a representative sample of the parent library. Class frequencies are therefore conditional on the stated MPO, SYBA, selectivity, multi-target, and scaffold-diversity filters and are not estimates for the full 65 856-molecule library.

Seventeen candidates were selected from the Set-C candidate panel generated from the library of 65 856 molecules described previously¹⁷. After $\text{MPO} \geq 0.70$, $\text{SYBA} > 0$, and $\text{SI} > 10$ filtering, 19 913 primary leads remained; the present Set-C cohort is a further prioritization of that filtered set. The candidate-selection panel is distinct from the production-MD cohort. Three sequential filters were applied: (1) MPO score ≥ 0.70 (primary leads); (2) SYBA score

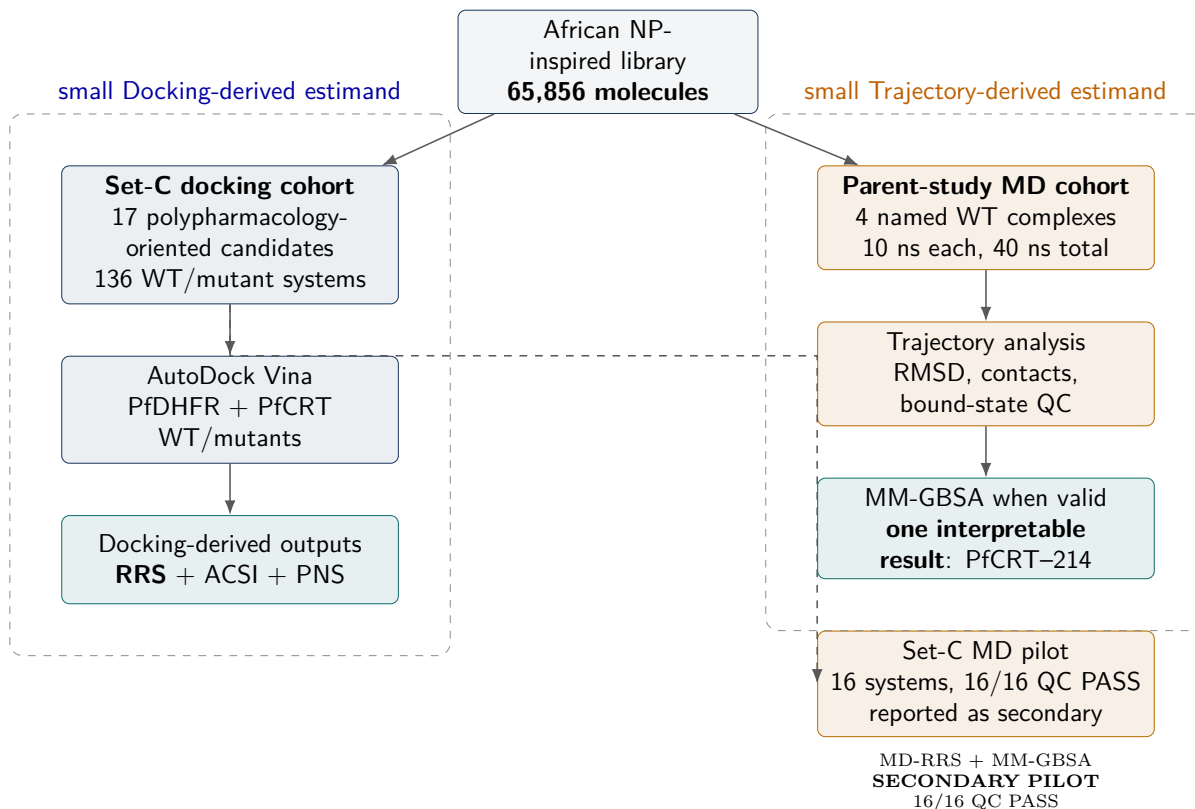


Figure 1: Study design. Docking-derived RRS, ACSI, and PNS are computed on the 17 Set-C candidates across 136 WT/mutant PfDHFR and PfCRT systems. Four parent-study complexes (non-overlapping with the Set-C list) undergo a single 10 ns production MD per system (CHARMM36m/GAFF2 or OpenFF/CHARMM36m for the parent-study and Set-C cohorts, respectively); a 16-system pilot (PP-01 and PP-02, single 10 ns replicate per mutant) is reported separately and does not define the RRS classes. None of the MD trajectories is replicated at the level recommended by the JCIM MD-reporting guidelines²¹.

> 0 (synthesisable); (3) selectivity index $SI > 10$ (selectively antiparasitic). The selection additionally required multi-target engagement of at least two of the four docking targets and scaffold diversity of no more than three compounds per Bemis–Murcko scaffold. The selection includes one of the previously named consensus leads (Ligand 87, PfDHFR). The four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were not part of the 17-candidate docking set but were used to construct the wild-type complexes for the production MD check. The 164 system is identified as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP, so this system is not structural validation of PfClpP. Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) were included as reference controls for the docking analysis (Table 1).

Table 1: Breakdown of the 136 docking protein–ligand systems by category (17 polypharmacological candidates $\times$ 8 wild-type/mutant PfDHFR and PfCRT structures). *Production MD was performed at 10 ns each (40 ns total) on 4 wild-type parent-study complexes distinct from the Set-C list. Ligand 164 is retained as the 164-labelled PfClpR cohort; PDB 4GM2 is PfClpR rather than PfClpP. The other parent-study systems were Ligand 201 (PfDHFR), Ligand 214 (PfCRT), and Ligand 438 (PfATP4).

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (Set-C)	17	5	85	—
PfCRT WT + 2 mutants (Set-C)	17	3	51	—
Docking total	—	—	136	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six resistance-associated states for comparison (Table 2); target identifiers are listed in the Supporting Information (??). Each state corresponds to a WHO-listed molecular marker of antimalarial drug resistance, giving the panel a clinical anchor rather than an arbitrary residue choice²².

Table 2: *Resistance mutation panel.*

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	PfCRT resistance/comparator state	Not a prevalence estimate

Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the six-mutant homology-model panel; moreover, PDB 4GM2 is PfClpR rather than PfClpP. Model quality was assessed by QMEAN score, Global Model Quality Estimation (GMQE), Ramachandran analysis (MolProbity), and C α RMSD to the wild-type template. Models were retained only when QMEAN exceeded -4.0 , GMQE exceeded 0.7 , Ramachandran favored residues exceeded 95 %, and backbone RMSD was below 1.5 Å. All six mutant models (PfDHFR N51I, C59R, S108N, I164L; PfCRT K76T, K76A) met these thresholds and were retained as docking receptors; a summary is given in the Supporting Information (??). K76A was included as a charge-neutralising isosteric perturbation of the canonical K76T substitution, allowing the contribution of the charged Lys76 anchor to be examined separately from the threonine substitution. Model scores are not experimental structures, and the lack of mutant crystal structures or replicate ensembles remains a source of uncertainty.

Docking protocol assessment

The docking protocol and V2-corrected grid parameters used here are identical to those established in the parent study¹⁷: AutoDock Vina 1.2.7¹⁵ with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ²³), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). As an independent boundary audit of these fixed boxes, P2Rank²⁴ pocket predictions were compared with the grid centers; the concordant PfCRT result and the frame caveats for the other receptors are presented in the Supporting Information (??).

Parent-study enrichment benchmarks (V2 grids) are summarized here and tabulated in full in the Supporting Information (??). On the MMV Malaria Box consensus (Vina+DiffDock), ROC-AUC was 0.924 (PfDHFR), 0.971 (PfCRT), and 1.000 (PfATP4). In contrast, DEKOIS 2.0 enrichment for PfDHFR with Vina alone was null (ROC-AUC 0.450, EF_{5%} = 0.00), indicating that score-based recovery of known actives is protocol- and library-dependent. Redocking was considered accurate when heavy-atom RMSD was < 2.0 Å relative to the crystal pose; the MTX redock attempt was a clear failure (RMSD ≈ 30 Å). AutoDock Vina was used for all reported Set-C scores; GNINA yielded no usable score for this panel.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO-affinity association was null and exploratory; complete correlations are provided in the Supporting Information (????).

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information

(????). These predictions were not experimentally validated.

Resistance resilience score definition

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. Grid boxes were centered on the co-crystallized ligand position with 25 Å dimensions, matching the audited V2 configuration files.

For target t , let $s_{\text{Vina},i,\text{WT},t}$ and $s_{\text{Vina},i,m,t}$ denote the signed Vina scores. Candidates with $|s_{\text{Vina},i,\text{WT},t}| < 5.0 \text{ kcal mol}^{-1}$ are excluded from that target’s score-retention analysis. The target-specific Resistance Resilience Score (RRS) is

$$\text{RRS}_{i,m,t} = \frac{|s_{\text{Vina},i,m,t}|}{|s_{\text{Vina},i,\text{WT},t}|} \times 100. \quad (1)$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or biochemical resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. To avoid letting unequal target coverage drive the main ranking, the primary analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are reported separately.

The mutually exclusive classes are defined from the available mutant values; primary class counts use the complete two-target subset:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least $7.0 \text{ kcal mol}^{-1}$.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below $7.0 \text{ kcal mol}^{-1}$.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below

80 %.

- **Class C:** at least one available mutant RRS value is at least 80 %, but not all available values reach 70 %.
- **Class D:** no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. The cut-offs follow the distribution of Vina scores in this panel and have not been calibrated to biochemical resistance or clinical outcome. Sensitivity to alternative thresholds is reported in the Supporting Information (??).

Secondary ranking descriptors

ACSI places each candidate relative to approved-drug and African natural-product reference spaces; weight-sensitivity checks are in the Supporting Information (??). PNS is a network-weighted ranking score; PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (??). Associations among metrics were assessed by Spearman rank correlation with 100 000 permutations, Bonferroni adjustment for three tests, and bootstrap intervals from 10 000 resamples. At $n = 12$ these correlations are descriptive and are not powered tests of independence.

MD protocol

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m²⁵/GAFF2²⁶ at 310.15 K using GROMACS^{27,28}. The 10-ns length is intended as a pose-retention and local-geometry check, not as a measurement of residence time or converged free energy. We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis²⁹ was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

MM-GBSA endpoint estimates were obtained from production trajectories using the GB^{OBC2} model (igb=5) with 0.150 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3¹⁸

CHARMM36-to-AMBER conversion. All MM-GBSA values are endpoint enthalpic scores from a single-trajectory protocol; configurational entropy is not included. Uncertainty is reported as the standard error of the mean over stored snapshots (101 for the interpretable PfCRT-214 system; 100 for Set-C pilot systems). Within-trajectory and propagated standard deviations are tabulated in the Supporting Information and are not replicate uncertainties.

The 16-system Set-C pilot used the same temperature and length, with ligand parameters from OpenFF 2.2.0 (AM1-BCC) rather than GAFF2. Absolute MM-GBSA values are therefore not compared between the two cohorts. Each mutant was run as a single replicate; trajectories were accepted for endpoint analysis only after geometric quality checks.

Results

Primary docking-RRS classification

Mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); full target-wise summaries are in the Supporting Information (??). The difference is in docking-score retention between panels, not evidence of different biological resistance.

On the complete two-target set ($n = 12$) the classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. When all 17 candidates are classified on whatever targets are eligible, the counts become A*: 5, A: 1, B: 5, C: 5, and D: 1; the five PfCRT-only compounds cannot be compared on equal footing with the two-target set. Per-compound RRS profiles across all six mutants are shown in ??. Varying the wild-type eligibility threshold from 4.0 kcal mol⁻¹ to 5.0 kcal mol⁻¹ left class counts unchanged; raising retention thresholds from 70 %/80 % to 80 %/90 % increased Class D from 1 to 9 (??).

Table 3: Target-specific docking-score retention for the 17 Set-C candidates. RRS is computed separately for each target and only when the corresponding wild-type Vina score magnitude is at least 5.0 kcal mol⁻¹. A dash denotes an ineligible target, not a zero score. The primary class applies only to the 12 complete two-target candidates; the available-target class is shown for all 17. Classes are mutually exclusive: A* requires all available mutant RRS values $\geq 80\%$ and the minimum eligible wild-type score magnitude ≥ 7.0 kcal mol⁻¹; A has the same RRS criterion but a weaker minimum wild-type score; B requires all values $\geq 70\%$ but not all $\geq 80\%$; C requires at least one value $\geq 80\%$ but not all values $\geq 70\%$; D is the residual class in which no available mutant reaches 80%. All scores are empirical AutoDock Vina scores; RRS is not a free-energy or resistance measurement.

Candidate	WT PfDHFR	WT PfCRT	Eligible targets	N51I	C59R	S108N	I164L	K76T	K76A	Primary class	Available class
PP-01	7.5	9.3	both	86.4	85.5	87.6	83.9	92.5	90.9	A*	A*
PP-02	–	9.1	PfCRT	–	–	–	–	87.9	94.6	–	A*
PP-03	10.3	11.3	both	68.1	68.3	65.8	69.0	80.0	81.4	C	C
PP-04	8.0	7.0	both	68.8	66.9	68.8	73.5	105.3	103.6	C	C
PP-05	–	10.5	PfCRT	–	–	–	–	77.1	79.3	–	B
PP-06	–	9.9	PfCRT	–	–	–	–	90.6	94.1	–	A*
PP-07	7.6	9.1	both	71.8	71.8	72.4	72.2	84.2	77.6	B	B
PP-08	7.6	7.4	both	62.1	62.1	67.2	66.3	83.9	83.1	C	C
PP-09	7.1	8.6	both	70.6	73.0	75.8	76.5	75.8	73.5	B	B
PP-10	7.7	8.4	both	76.0	76.4	75.7	75.8	89.4	89.6	B	B
PP-11	–	10.0	PfCRT	–	–	–	–	82.7	82.5	–	A*
PP-12	7.5	7.8	both	75.2	66.1	62.9	72.5	70.4	87.2	C	C
PP-13	–	8.3	PfCRT	–	–	–	–	110.2	99.4	–	A*
PP-14	7.6	7.5	both	62.2	65.5	68.0	64.9	78.8	77.7	D	D
PP-15	5.6	9.3	both	123.2	112.7	125.9	131.8	86.2	90.1	A	A
PP-16	7.6	8.7	both	72.9	74.7	73.7	75.9	80.1	80.6	B	B
PP-17	8.3	7.4	both	59.3	61.2	59.5	58.9	76.8	93.4	C	C

Chemical-space and network descriptors

Two of 17 candidates had ACSI above 0.70 (cohort mean 0.543). ACSI measures proximity to reference chemical space, not activity or resistance. PNS rankings and imputation details are in the Supporting Information (??).

Cross-metric associations

Table 4: Cross-metric Spearman correlations on the complete two-target set ($n = 12$). Permutation p -values use 100 000 permutations; adjusted p -values use Bonferroni correction for three tests. Bootstrap intervals are percentile 95% intervals from 10 000 resamples. At this sample size the correlations are descriptive.

Metric pair	Hypothesis	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95% interval
PNS vs. RRS	H1	−0.209 8	0.514 4	1.000 0	[−0.7582, 0.5429]
ACSI vs. RRS	H2	−0.405 6	0.192 2	0.576 6	[−0.8917, 0.2857]
RRS vs. weakest WT score	H3	−0.166 1	0.603 8	1.000 0	[−0.7300, 0.5932]

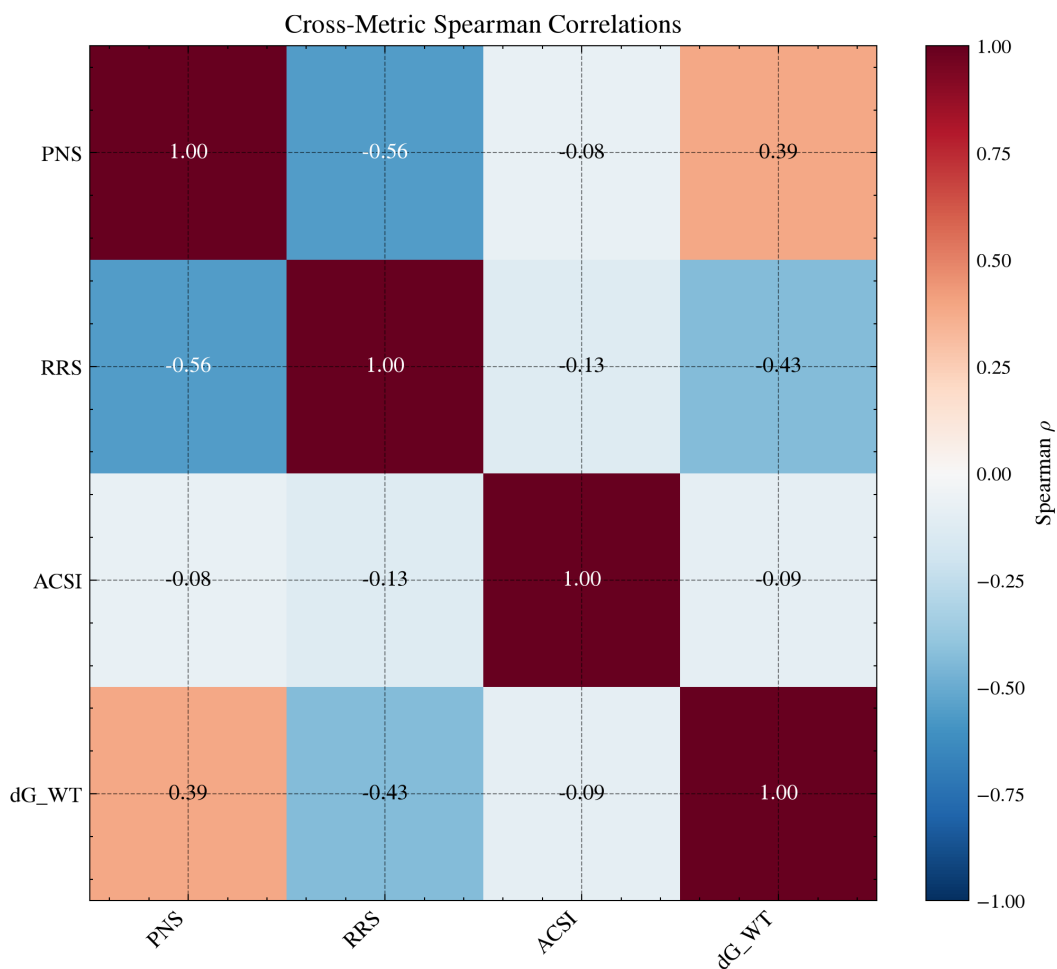


Figure 2: Cross-metric Spearman correlations for Set-C candidates. The primary analysis uses the 12 complete two-target candidates; the 17-candidate matrix is shown only for comparison when target coverage is incomplete. PNS already includes wild-type docking scores, so that pair is not an independent test.

Table 4 reports the complete two-target set. On all 17 candidates, PNS-RRS was $\rho = -0.5588$ (permutation $p = 0.0222$, adjusted $p = 0.0667$), ACSI-RRS $\rho = -0.1324$ ($p = 0.6117$), and RRS versus weakest wild-type score $\rho = 0.1759$ ($p = 0.4947$), but unequal target coverage limits interpretation of the full set.

On the two-target set, PNS and RRS were only weakly associated ($\rho = -0.2098$, adjusted $p = 1.0000$), as were ACSI and RRS ($\rho = -0.4056$, adjusted $p = 0.5766$). RRS was not associated with the weakest eligible wild-type score ($\rho = -0.1661$, adjusted $p = 1.0000$). PNS already includes the wild-type Vina scores that also enter the RRS computation, so the PNS-RRS pair shares an input and is not an independent test. The bootstrap intervals are wide, as expected for $n = 12$.

Computational prioritization intersection

Requiring Class A or A*, ACSI > 0.5 , and above-mean PNS leaves PP-15 on the two-target panel; PP-15 was, however, included only as a feasibility probe outside the 17-candidate Set-C estimand (see ?? ; the PP-15 feasibility details are provided in the Supporting Information) and so is not a product of the Set-C prioritisation pipeline. That intersection is reported as a shortlist for experiment, not as evidence of activity or resistance resilience.

Parent-study MD check (non-overlapping cohort)

These four complexes test whether docked poses remain bound on a short timescale; they do not validate the Set-C ranking (??).

Table 5: Production MD stability metrics for parent-study wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

Two systems, PfCRT and PfATP4, retained their ligands throughout production. PfCRT

had a backbone RMSD of 4.47 Å, a ligand RMSD of 2.45 Å, 78.1 contacts, and 23.1 H-bonds (minimum heavy-atom distance 3.19 Å). PfATP4 shows a backbone RMSD of 12.27 Å, a ligand RMSD of 2.13 Å, 178.4 contacts, and 47.7 H-bonds (minimum distance 2.25 Å). The high backbone RMSD indicates substantial structural drift in this model. Because the simulation omitted a membrane and the topology conversion was not valid for MM-GBSA, the apparent ligand persistence should not be interpreted as validated binding thermodynamics. PfCRT ligand 214 anchors to Lys34 and engages in a stable H-bond network (Gln101, Thr30, Gln297) and a hydrophobic core (Leu301, Leu105, Ala33), consistent with a persistent pose in this trajectory.

The other two systems, the 164-labelled PfClpR cohort and PfDHFR, had stable protein conformations (backbone RMSD 1.31 Å and 3.05 Å, respectively) but the ligands are completely unbound (minimum distances 67.42 Å and 78.21 Å, zero contacts). This indicates either incorrect initial placement or rapid dissociation during equilibration, not a PBC artefact. Consequently, MM-GBSA analyses for these systems are not interpretable as affinity estimates. Protein equilibration alone was insufficient to retain the docked poses.

Table 6: Endpoint interpretation for four parent-study complexes used for production MD. Only PfCRT-214 yields an interpretable MM-GBSA endpoint; dissociated systems are reported as N/A.

Ligand	Target	ΔG_{bind} (kcal mol ⁻¹)	SEM	Interpretation
214	PfCRT-WT	-18.25	0.40	Interpretable; ligand bound [†]
164	PfClpR-labelled cohort-WT	—	—	N/A; ligand dissociated [‡]
201	PfDHFR-WT	—	—	N/A; ligand dissociated [§]
438	PfATP4-WT	—	—	N/A; conversion corrupted*

[†]Ligand remains bound (minimum distance 3.19 Å, 78 contacts); conversion via `gmx_MMPBSA` native converter.

The quoted uncertainty is the standard error of the mean over the 101 stored snapshots (see Methods).

[‡]Ligand dissociates to 67.4 Å minimum distance with zero contacts during production; excluded from binding-mode conclusions.

[§]Ligand dissociates to 78.2 Å minimum distance with zero contacts; same caveat applies.

*Excluded: conversion artifact in the two-chain PfATP4 topology (+473 kcal mol⁻¹ van der Waals inflation); structural persistence is reported separately.

Only the PfCRT-ligand 214 system yields an interpretable MM-GBSA endpoint esti-

Figure 3: Detailed stability analysis for the parent-study PfCRT–Ligand 214 complex over 10 ns production MD. The system exhibits transporter flexibility while retaining a bound ligand; cohort-level stability values are reported in Table 5.

mate ($\Delta G_{\text{bind}} = (-18.25 \pm 0.40) \text{ kcal mol}^{-1}$). The 164-labelled and PfDHFR-201 ligands dissociated during production; the PfATP4 system was excluded because CHARMM36-to-AMBER conversion of its two-chain topology produced a systematic van der Waals artifact ($+473 \text{ kcal mol}^{-1}$). No Vina-MM-GBSA correlation is estimated across the parent-study and Set-C cohorts.

Set-C MD pilot (secondary)

Docking-RRS and short-MD structural retention are non-equivalent estimands

The primary and secondary quantities are summarized in the Supporting Information (??????).

A 16-system pilot on PP-01 and PP-02 asked whether short trajectories and MM-GBSA would recover the docking-RRS pattern. All 16 trajectories met the geometric acceptance checks. For the eight mutants with a docking wild-type reference, docking predicted weaker mutant scores, while trajectory metrics more often indicated retained or tighter contacts (seven of eight comparisons disagreed in direction). This divergence is the principal calibration result: docking scores describe a static empirical scoring comparison, whereas the short MD metrics describe local geometric retention over the tested trajectory. Neither measurement is a direct estimate of affinity, residence time, target engagement, or resistance. MM-GBSA mutant/wild-type ratios exceeded 100 % for eight of twelve mutants; those ratios are endpoint comparisons, not free-energy measures of resilience. The pilot therefore does not reproduce docking-RRS and does not imply resistance or tighter binding. Full values and the threshold-margin sensitivity analysis are in the Supporting Information (??????).

As a bounded consistency check, we intersected the docking-RRS retention criterion (class-A, $\text{RRS} \geq 80\%$) with pilot MD-RRS retention parity (mean minimum-distance ratio < 100) at pilot scope. Seven of twelve gate-eligible rows passed both gates: PP-01 satisfied

the two-target gate (PfCRT via K76A, PfDHFR in four of four mutants), whereas PP-02 did not, because its PfDHFR states lack a docking wild-type anchor under the 5.0 kcal mol⁻¹ eligibility rule and only one target was gate-eligible. The single row above parity (PP-01 PfCRT K76T, 102.7) lies within the 2.0 kcal mol⁻¹ endpoint noise floor and is read as retention, not gain (??).

A single-ligand feasibility probe (PP-15, outside the Set-C cohort) extended the same OpenFF protocol to wild-type PfDHFR and PfCRT: Vina best poses were -8.62 kcal mol⁻¹ and -7.81 kcal mol⁻¹, respectively, and both systems completed 10 ns production trajectories with valid checkpoints, yielding MM-GBSA estimates of (-29.52 ± 0.33) kcal mol⁻¹ (PfDHFR) and (-27.79 ± 0.49) kcal mol⁻¹ (PfCRT); as for the pilot, no RRS estimate is reported for this probe (??).

Table 7: Set-C pilot MM-GBSA estimates (exploratory; not used to assign RRS classes). ΔG_{bind} is the mean over 100 snapshots ± SEM; within-trajectory standard deviations are in the Supporting Information. MMG-RRS is the mutant/wild-type endpoint ratio (×100; wild type = 100). Docking RRS is shown for comparison (n/a: no wild-type docking reference for PP-02 PfDHFR). Single-replicate endpoints should be read only as within-protocol contrasts.

Candidate	Target	State	ΔG_{bind} (kcal mol ⁻¹)	SEM	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	—
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	—
PP-02	PfCRT	K76A	-24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	—
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	-30.16	2.06	100.0	—

External docking replication

Docking of 39 external ligands across the same eight PfDHFR/PfCRT states gave 312 finite scores. Bootstrap mean RRS was 100.45 (99.59–101.32) and the Class-A fraction was 0.974 (0.923–1.000). A mean near 100 means mutant and wild-type Vina scores were on average

almost proportional, so this panel adds little mutant/wild-type contrast under the same protocol. It is a computational sensitivity check, not validation of the Set-C ranking or of experimental resistance (??).

Discussion

Docking-derived triage rather than validated resilience

RRS compares each mutant score with the corresponding wild-type score after excluding wild-type magnitudes below $5.0 \text{ kcal mol}^{-1}$. It is a docking-derived prioritisation statistic, not a biochemical measurement of resistance or affinity. Class A* combines a wild-type score magnitude of at least $7.0 \text{ kcal mol}^{-1}$ with RRS values of at least 80 % across all tested mutants; Class A applies the same 80 % retention threshold with a weaker wild-type magnitude floor; Class B requires all available mutants above 70 % with at least one below 80 %; Class C requires at least one mutant above 80 % but not all above 70 %; and Class D denotes that no available mutant reaches 80 %. The classes are mutually exclusive, combinatorially defined, and not evidence of resistance circumvention. They can be aligned with regional resistance surveillance in a future decision analysis, but they do not justify clinical deployment or therapeutic recommendations on their own³⁰.

This distinction is relevant in the current epidemiological context. The 2025 World Malaria Report documents declining efficacy of some artemisinin partner drugs alongside the spread of *Pfkelch13*-mediated partial resistance, with both trends concentrated in the African Region¹. Resistance-aware prioritization is a useful design criterion, though not a guarantee of clinical relevance: compounds selected without an explicit mutant-sensitivity analysis are poorly matched to the resistance states that are relevant in a given parasite population, but computational scores alone cannot establish that mismatch or its therapeutic consequence.

The external docking panel (mean RRS ≈ 100) underscores the same boundary. Under

identical scoring and mutation definitions, mutant and wild-type Vina scores remained nearly proportional on an independent ligand set. Possible explanations include limited sensitivity of the Vina score to these particular mutations in that chemical space, insufficient structural perturbation captured by homology models, or genuine score retention for many of those ligands. The data cannot distinguish among these alternatives; they caution against treating RRS as a calibrated resistance predictor outside the primary cohort.

A same-pose cross-scoring sensitivity analysis addresses one of these alternatives. Rescoring the same 312 poses with the GNINA CNN affinity (post-processing only) reproduced the class-level picture: all 38 eligible ligands were class A under both scoring layers, with no class discordance (??). The retention-not-gain pattern is therefore not an artefact of a single scoring function. The ligand-level rank correlation between layers was moderate ($\rho = 0.558$) and per-mutant correlations were weak or null (e.g. C59R -0.076), so the consensus supports class-level retention but not per-mutant rank transfer across scoring functions. Bootstrap intervals (1×10^4 resamples) bound the same conclusion on the primary cohort: the class- A* fraction was 0.294 (0.118–0.529) and the PfCRT per-mutant retention intervals excluded 100 (K76T 85.4, 81.0–90.4; K76A 87.0, 83.1–90.9), whereas the PfDHFR intervals were wide and overlapped 100 for I164L (??).

The relationship between RRS and molecular topology remains unresolved. The companion TDA analysis reported associations in separate overlapping cohorts³¹; because those candidates share provenance with the present study, the estimates are not independent validation. Topological descriptors should therefore be tested prospectively in larger, matched panels with mutant-state free-energy or experimental measurements before being used as a design rule.

The PfCRT component of this picture is mechanistically plausible in light of long-timescale simulations of the chloroquine resistance transporter: 130 μ s of atomistic MD across CQ-sensitive and CQ-resistant isoforms established that Lys76 physically obstructs the drug-binding site while serving as the electrostatic anchor for peptide substrates, with R371I and

Q271E acting in concert with K76T to open drug access³². Our mutant-vs-wild-type endpoint contrasts are consistent with this charged-cavity picture, in which perturbing residue 76 reshapes the local electrostatics that gate both ligand retention and substrate specificity. This consistency is contextual support, not an independent validation of the present docking scores.

Chemical and network context

ACSI places candidates relative to approved-drug and African natural-product chemical space. The mean was 0.543, and two of 17 candidates (11.8%) exceeded 0.70. PNS is a network-weighted rank; on the two-target set it associated only weakly with RRS ($\rho = -0.2098$, adjusted $p = 1.0000$). At $n = 12$ the correlations mainly describe how the ranks line up in this cohort; the wide bootstrap intervals confirm that the three ranking dimensions capture distinct information. ACSI, PNS, and RRS are complementary ranking tools, not independent measures of biology, and their meaning depends on the chosen reference spaces, network, and targets^{11,13,14}.

Role of the structural stress test

Short MD on the parent-study complexes shows both the value and the limit of structural follow-up when the compounds are not the ranked Set-C set. Two of four systems kept a bound ligand; the 164-labelled PfClpR and PfDHFR-201 systems did not. Only PfCRT-214 passed the checks needed for MM-GBSA, giving (-18.25 ± 0.40) kcal mol⁻¹—a system-specific endpoint that cannot calibrate Set-C scores. As is standard for endpoint methods, that number is a within-protocol guide, not a converged free energy^{18,19}.

The Set-C pilot is the only trajectory comparison with docking-RRS for compounds that are actually on the prioritization list. Its central contribution is calibration of interpretation: it tests whether a static score-retention signal and short-timescale structural retention behave as interchangeable quantities, rather than attempting to validate either quantity as

a biological endpoint. Mutant MM-GBSA did not recover the docking-predicted weakening, so whether RRS tracks mutant-state free-energy differences remains unresolved. With one replicate and 10 ns per system, the pilot addresses local geometry and endpoint consistency, not residence time or experimental resistance. The pilot-scope retention gate illustrates the same boundary at class level: PP-01, the only candidate with a docking wild-type anchor on both targets, satisfied the two-target gate (docking RRS $\geq 80\%$ together with pilot MD-RRS below parity), whereas the directional mismatch observed in most mutant–wild-type pairs persists. Because the pilot bound fraction is saturated at 100% for all sixteen systems, the gate is a coarse consistency check, not an affinity-equality proof.

Non-equivalence of docking-RRS and MD-RRS estimands

The central question of this study asked whether docking-RRS and MD-RRS measure the same quantity. The answer is clearly negative: seven of eight mutant comparisons showed directional disagreement. This non-equivalence is not a failure of either method but a consequence of what each estimates—docking-RRS reflects a static score ratio between mutant and wild-type poses, while MD-RRS captures dynamic binding persistence under explicit solvent and protein fluctuations. The two quantities diverge when conformational selection or induced-fit effects alter the binding mode upon mutation, which is precisely the regime relevant to drug resistance. For antimalarial lead prioritization, this means that docking-based RRS serves as a first-pass computational filter to stratify candidates, but it is not a validated prediction of resistance resilience. MD stress tests, even short ones, provide a complementary but non-redundant signal that probes a different aspect of the binding event. The practical consequence is that computational prioritization and structural scrutiny should be reported as parallel ranking dimensions with non-overlapping failure modes, not as convergent evidence of resistance resilience.

What would constitute validation. None of the scores reported here is biochemical or biophysical evidence. The most direct test would be matched IC₅₀ measurements of the

highest-RRS candidates on wild-type and mutant PfDHFR; PfCRT binding or transport assays at digestive-vacuole pH; and short growth-inhibition tests in culture to determine whether multi-target computational signals translate to a measurable phenotype. Agreement or disagreement with experiment would identify which computational signal, if any, is useful for each target. The multi-target, mutation-resilient candidates prioritised here are also natural inputs to machine-learning predictors of antimalarial combination synergy, in which the same molecular features would be scored for pairwise combination effects³³; such synergy predictions remain outside the scope of the present study.

Comparison with existing polypharmacology and resistance-aware approaches

Conventional multi-target docking ranks compounds by aggregated wild-type scores and does not treat mutant-state retention as a primary quantity. Network pharmacology often weights targets by centrality without comparing wild-type and resistance-associated structures under one scoring protocol¹¹. Under the SDMT frame, the present class-A* readout combines exactly the two axes such a strategy requires—target breadth in the four-target wild-type matrix and per-target mutational retention in the RRS panel—and is therefore reported as an SDMT-style hypothesis, not as demonstrated multi-target engagement⁹. Free-energy methods such as FEP or TI can report mutant binding differences with clearer physical content, but they remain costly for early triage and need reliable structures and force fields. The approach used here sits between those extremes: RRS is an explicit mutant/wild-type docking-score ratio, while ACSI, PNS, and short MD remain separate layers rather than substitutes for one another. None of them replaces IC₅₀ panels, transporter assays, or converged free-energy calculations; they are intended only to shortlist compounds for those experiments^{20,31}.

Limitations

Mutant receptors were homology models. All six met the QMEAN, GMQE, Ramachandran, and RMSD thresholds, yet model quality is not experimental accuracy. MD was limited to 10 ns on four parent-study complexes and a single-replicate 16-system Set-C pilot; neither design addresses long residence times or convergence. The JCIM MD-reporting guidelines²¹ recommend at least three replicates per system with adequate duration and convergence; no new replicate campaign was launched for the present submission, and the pilot is therefore reported strictly as a secondary protocol-level structural stress test rather than as a thermodynamic or kinetic measurement. An independent PP-01 PfCRT wild-type run differed from the reference endpoint by $2.01 \text{ kcal mol}^{-1}$, which we take as a rough noise floor given snapshot autocorrelation (??). Single-replicate mutant endpoints, including the Set-C MM-GBSA values, are sensitive to frame choice and periodic-boundary handling. MM-GBSA omits configurational entropy; only PfCRT-214 satisfied the bound-state and topology requirements. The four-target panel is not a full parasite targetome.

RRS class frequencies depend on the selection filters and on unequal target coverage; they should not be extrapolated to the full library. The cut-offs follow the distribution of Vina scores in this panel and have not been calibrated to biochemical resistance or clinical outcome (??). PfCRT network centrality was imputed, so PNS depends on how the network is built (??). Network-pharmacology scores are generally sensitive to database coverage and hub bias³⁴; the PNS reported here is therefore a bounded ranking heuristic, not a validated network property. Force fields differed between the parent-study and Set-C cohorts, so absolute MM-GBSA values are not compared across protocols. Cross-metric correlations rest on $n = 12$ complete two-target candidates; the wide bootstrap intervals **confirm** that the three ranking dimensions capture distinct information rather than redundant signals. PfCRT was prepared at physiological pH rather than digestive-vacuole pH, which can change rankings¹⁷. Five candidates lacked an eligible PfDHFR wild-type score and appear only in the PfCRT-only analysis, a consequence of receptor setup rather than demonstrated selectivity.

Conclusion

Resistance-aware polypharmacology docking produces a discriminant prioritization signal inside the curated 17-candidate cohort: the 12 candidates with complete PfDHFR and PfCRT data fall into five mutually exclusive classes (A*–D) that separate predicted wild-type potency from predicted mutant retention (??). The three ranking dimensions—RRS, PNS, and ACSI—capture distinct information, with weak cross-correlations confirming their complementarity. Short molecular dynamics stress tests probe a different estimand and cannot serve as direct validation of docking-derived retention classes: on non-overlapping parent-study complexes, ligand retention occurred in only two of four systems; on the Set-C pilot, trajectory metrics and docking-RRS disagreed in direction for seven of eight mutant comparisons. Docking-RRS and MD-RRS are non-equivalent quantities; computational prioritization and structural scrutiny should be reported as parallel ranking dimensions with non-overlapping failure modes, not as convergent evidence of resistance resilience. An external docking panel averaged near RRS 100, with little mutant/wild-type contrast under the same protocol. The ranking is useful inside this computational panel as a hypothesis-generating prioritization; it does not constitute a validated triage procedure and does not demonstrate biological polypharmacology, a systems-level mechanism, or experimental resistance circumvention. PP-01 **merits** experimental follow-up; PP-02 was not promoted by the pilot gate. Target-engagement assays, phenotypic tests, longer or free-energy simulations on mutants, and prospective experiments remain necessary before any biological claim can be advanced.

Authors’ Contributions

Myke Vital Sao Temgoua: Methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapet Njafa: Methodology, software, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review &

editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. Serge Guy Nana Engo: conceptualization, methodology, supervision, writing – review & editing. All authors read and approved the final manuscript.

Data Availability

Docking results, metric outputs, analysis scripts, and the associated source-data records are available at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. A versioned snapshot of this repository will additionally be archived on Zenodo to provide a permanent, citable DOI concurrent with publication. The large parent-study trajectories are not included in this article’s data record; the conclusions drawn here are restricted to the trajectory-derived summaries and binding estimates explicitly reported above.

Competing Interests

The authors declare no competing interests.

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Use of Artificial Intelligence

AI assistants were used to help with code and analysis scripts. All four Springer Nature AI-policy expectations were respected: (i) human accountability for every scientific claim remains with the listed authors; (ii) AI supported but did not replace scholarly judgement,

editorial decision-making or the design of the comparison; (iii) the use of AI is disclosed here transparently; (iv) no confidential manuscript material, peer-review content or restricted data were shared with external AI services. All AI-assisted text and code were reviewed and edited by the authors. No generative AI tool is an author. Computational results, statistics, and quantitative claims were produced and checked by the authors.

Supporting Information Available

Docking enrichment and homology-model quality tables; P2Rank box audit; MPO, ADMET, and ACSI sensitivity; RRS radar profiles and threshold sensitivity; PNS imputation and Tartarus calibration; cohort and evidence-scope tables; PlasmoDB identifiers; parent-study and Set-C pilot MD/MM-GBSA summaries; robustness and Paper 1 context tables (PDF).

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TOC Graphic

